

To: Department of Health and Human Services

From: PictorLabs

Date: Jan 20, 2026

RE: Request for Information: Accelerating the Adoption and Use of Artificial Intelligence as part of Clinical Care [RIN 0955-AA13]

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Pictor Labs is a virtual staining company advocating strongly for digital pathology adoption through Artificial Intelligence (AI) technology that delivers results in minutes while preserving tissue for additional and often required comprehensive analyses. We applaud the Department of Health and Human Services (HHS) on its initiative to learn more about the impact of AI in healthcare and the influence of reimbursement and payment policy on its access to clinical care.

AI has quickly evolved to allow for the development of critical tools in healthcare. Of importance to our company, AI increases the efficiency, and thus improves, the normal laboratory workflow of staining histological slides for medical assessment. These tissues can be virtually stained with routine biomedical stains (e.g., H&E), special tissue stains such as Masson's Trichrome, or immunohistochemical stains which selectively identify a single molecule of interest.

This beginning step in the analysis of tissue biopsies/needle aspirations is critical for the ultimate diagnosis and/or decision to have additional stains performed as part of a differential or for confirming diagnosis, and yet becomes a bottleneck to *all* subsequent procedures due to the time involved for irreversible chemical staining. **PictorLabs believes that the HHS should prioritize the first pivotal steps that many healthcare decisions rely upon. Tissue staining is one such primary step on which the pathologist diagnosis depends, and a fundamental step that AI processes can make more time-effective for the patient by circumventing delays in diagnosis.** Importantly, making this primary step available to laboratories and ultimately physicians can also reduce disparities in healthcare access and equity for patients, as the availability of pathology laboratories capable of a myriad of staining may be functionally or locally limited, or financially prohibitive.

There are also added benefits for urging AI enabled healthcare technology in the assertion of American autonomy. Digital staining, as an example, can reduce economic dependence on other countries. Currently, laboratories in the US depend on obtaining traditional staining chemicals from other countries, such as China and India. Finished kits may be bottled in the US/EU, but upstream dyes/intermediates frequently originate abroad—so shocks in those supply chains quickly reach and affect US labs logistically with procuring the stains as well as financially with altered pricing.

Therefore, the ability of US laboratories to confidently maintain the ability to virtually stain and keep their workflow moving is a positive alternative to the dependence on other supply chains.

In order to expedite the advancement of AI in healthcare, we would seek assistance from HHS in certain areas that include:

- ❖ A defined and consistent pathway to reimbursement that is flexible for many scenarios. Current approaches for coding and coverage determination move more slowly than the advancement of technology, and thus hampers confidence in reimbursement that is necessary for bringing innovation to market
- ❖ Changing the inertia of the healthcare system to utilize AI tools to increase efficiency and output of healthcare staff through education on new technologies

Current avenues for reimbursement, such as the New Technology Add-on Payment (NTAP), exist but are technology-specific and involve a complex approval process. Potential alternatives include a reimbursement framework that proposes an analytical pathway, similar to the one successfully employed for AI-driven diabetic retinal exams (see <https://pubmed.ncbi.nlm.nih.gov/35681002/>). While meritorious, it too involves extensive analysis for each and every step that exists to utilize the technique and cannot be incorporated quickly enough to incentivize. Existing legislation, such as the Health Tech Investment Act (S. 1399), aims to assign Advanced Health Behavior Services (AHBS) to a new technology ambulatory payment classification (APC). This legislation has garnered support from AdvaMed, the representative body for the medical biotechnology sector (see <https://www.advamed.org/industry-updates/news/advamed-applauds-house-introduction-of-the-health-tech-investment-act/>). Currently, this legislation remains waiting for formal discussion.

A simple first step of establishing identical reimbursement for current staining procedures (e.g., CLIA fees) would greatly attenuate hesitancy by a pathology laboratory to adopt AI-driven technology. If a stained slide is reimbursed, whether or not that is accomplished by approved AI or chemical processes should be unimportant. Both AI and traditional staining require an investment of time and resources - only for AI staining the time expenditure is upfront in development and validation. As such, it is our opinion the stain should be reimbursed equally. But there is also an important overlooked benefit from virtual staining in that virtual stains require only a single tissue slice and thus a significant amount of biopsy tissue is retained. This tissue retention could be a critical issue for patients, as additional molecular tests are frequently needed for diagnosis confirmation and therapy guidance. Tissue retention also impacts care, as several common biopsy locations (e.g., breast, melanoma, colorectal, kidney, lung, thyroid) are frequently inadequate and commonly trigger repeat sampling and follow-up. Repeat biopsies are a real, recurring source of healthcare cost and patient burden. If a virtually stained slide circumvents a re-biopsy, significant cost is saved, not to mention sparing the patient another invasive procedure, delays, frustration, and anxiety. Therefore, prioritizing this primary staining step ensures that the benefits of digital staining will be multifactorial, more efficient, and not restricted to a single demographic or therapeutic area.

Education presents another critical challenge for AI-based companies. The United States and Canadian Academy of Pathology (USCAP) and the Digital Pathology Association (DPA) have both acknowledged the potential of this technology and the necessity for its advancement. Both societies currently set a compelling example through collaboration with AI and digital pathology companies, thereby demonstrating a commitment to adopting cutting-edge technology that enhances the quality and speed of diagnostics. For many years, the DPA has worked with the FDA to help secure clearances for whole slide imaging equipment, and thus has created a more productive workplace for pathologists and healthcare institutions as well as enhancing patient care. Thus, medical education towards AI digital pathology is timely, recognized, supported by USCAP and DPA, and provides a successful example of regulating bodies co-operating with such societies for continued education and advancement of AI efforts. Subsequent intensive interaction for HHS and USCAP/DPA here is suggested.

A prevalent viewpoint suggests that a central source of reluctance to utilize novel AI technology stems from "the black box nature of AI" (see <https://pubmed.ncbi.nlm.nih.gov/40140300/>; <https://pubmed.ncbi.nlm.nih.gov/37093631/>). Governing bodies can also provide significant assistance in the adoption of new technology. Support for medical education can be accomplished by regulating bodies through supporting CME, the inclusion of modern technology use for hospital accreditation, and the establishment of required insurance coverage for AI-dependent processes that aligns effectively with reimbursement strategies could ensure more rapid clinical decisions impacting patient care, as well as ensure the healthcare system stays current with important technological progress.

In summary, HHS should prioritize foundational and simple healthcare steps such as virtual histopathological staining for AI-driven advancement. The adoption, reimbursement, and acceptance of AI-driven processes is critical to the continued successes our healthcare system brings to patient care. While we highlight policy needs related to our specific business, those principles are applicable across all healthcare companies innovating with AI. Thank you for the opportunity to comment on this important area of attention.